

PRODUCTIVITY & CHROME EXTENSION PLATFORM

Project Overview

Student productivity platform blocking distracting websites, featuring YouTube Study Mode, focus session timers, streaks & gamified leaderboards.

Problem Statement:	Students lose study hours to social media notifications and algorithmic YouTube recommendations during critical study sessions.
Solution Implemented:	Built a Chrome Manifest V3 extension & web platform enforcing scheduled website blocks, YouTube feed removal, and focus telemetry.
My Technical Role:	Lead Full-Stack Developer & Chrome Extension Architect

Main Features & Technical Stack

Core Features:	Technologies Used:
- Scheduled site blocklist & custom block rules - YouTube Study Mode stripping recommended video feeds - Focus Pomodoro timer with automated history logging - Gamified student focus streaks and leaderboards	React, Vite, FastAPI, Python, SQLite, SQLAlchemy, Chrome Manifest V3, Vercel
	Architecture Flow:
	React / Extension Popup → FastAPI (Python) API Gateway → JWT Auth → SQLite DB → Analytics Engine

Technical Challenge & Results

Key Engineering Challenge: Manifest V3 background worker lifecycle termination fixed using persistent Alarm APIs & offscreen documents.

Validated Results: Proven zero false-positive site blocks and hundreds of recorded deep-work focus hours.

Live Demo: <https://github.com/prashant-singh-78/SWD.git> | **GitHub:** <https://github.com/prashant-singh-78/SWD.git> | **Portfolio:** prashant08.vercel.app